

PR-AYC-G / Topo-OSM Working Manuscript

State-locked narrative meaning, sensory entrainment, decoder-gate availability, and update-polarity modeling

v0.5 - DGA / Phase-Scrambled Control / Wild-Hare Boundary Update

Private pre-submission working draft - exploratory methods and hypothesis architecture only

Manuscript status and claim boundary

This draft is not a submission-ready paper. It summarizes a developing protocol, software stack, synthetic identifiability work, and single-participant exploratory self-runs. PR-AYC-G is presented as exploratory psychophysiology. Topo-OSM is a cautious human-scale latent-state interpretation layer. Current evidence does not prove consciousness, cellular OSM biology, microtubular memory, biophotonic memory, clinical efficacy, metaphysical truth, or literal thermodynamic energy transfer.

Abstract

The PR-AYC-G protocol was developed to ask whether accessible narrative meaning produces a measurable body-brain state trajectory that differs from matched sensory stimulation and from the same intact narrative under analytic task stance. The current architecture compares phase-scrambled sensory-control, intact Target, and Contextual Override conditions, recorded with EEG, autonomic streams, event markers, stimulus-fingerprint regressors, self-report, and confound reports. Since v0.4, the framework has been compressed around A-MRED and MRED-Peak / MRED-Resolution as the primary interpretive endpoint family, while adding Decoder Gate Availability (DGA) as a secondary interpretive layer. DGA formalizes a constraint exposed by the Eternal Sunshine run: semantic access, emotional absorption, and integration can dissociate across branches when sensory clarity and receptive stance are not aligned. The updated manuscript also reframes the phase-scrambled branch as a meaning-damaged sensory entrainment control rather than a meaningless condition. Across Arrival, Contact, Field of Dreams, and Eternal Sunshine, the most defensible current claim is not that meaning has been proven as a physical force, but that PR-AYC-G now provides an auditable method for asking whether meaning recognition, integration, update polarity, and gate availability have distinguishable temporal structure.

Contribution map - updated claim discipline

Layer	Permitted current claim	Not permitted from current evidence
PR-AYC-G protocol	An exploratory three-arm psychophysiology design comparing sensory-control, intact Target, and analytic Override.	Does not prove consciousness, soul, love as force, OSM, or clinical effect.
Phase-scrambled Control	A meaning-damaged sensory entrainment branch designed to preserve audiovisual/cue energy while damaging recognizable narrative meaning.	Not a physiologically dead or absolutely meaningless branch.
A-MRED / MRED-Peak	A future primary endpoint: locked anchor shows Target-specific recognition plus delayed integration after QC and CET-R gates.	Not valid when anchors are estimated, ALS fails, or confounds dominate.
MRED-Resolution	A delayed after-state / recovery endpoint for closure-like narratives	Does not prove healing, therapy, or memory formation.

	where the effect lives in Baseline2 or washout rather than sharp peaks.	
DGA	A secondary model of semantic access, decoder availability, emotional impact, extraction load, and confound burden.	Does not prove an innate metaphysical decoder; it estimates gate conditions.
Topo-OSM	A cautious human-scale network-state/topology interpretation layer.	Not cellular hidden-Y, microtubules, or biophotons.
Opto-PING Rung 0	Synthetic identifiability: the software can detect a known simulated reciprocal loop under locked assumptions.	Not biological proof.

1. Updated core hypothesis

Current core sentence

PR-AYC-G tests whether meaning has timing, shape, physiological consequence, and access conditions - not whether meaning can be reduced to any one signal.

The updated hypothesis has four separable claims:

1. Meaning recognition and meaning integration can dissociate.
2. Acute meaning peaks and delayed resolution/recovery profiles can have different physiological shapes.
3. Analytic extraction does not simply erase meaning; it can reroute, delay, or alter the way meaning is accessed.
4. Meaning appears to require both an external key and an internal decoder: sensory access, prior state-space, personal relevance, autonomic availability, and low enough task/confound burden.

Meaning response != f(stimulus only)

Meaning response ~ f(stimulus geometry, sensory access, prior state-space, personal relevance, autonomic availability, extraction load, confound burden)

2. Phase-scrambled Control is not optional for serious PR-AYC-G

The phase-scrambled branch should remain in any serious PR-AYC-G run. It is the defense against the exogenous entrainment critique: the possibility that the brain is only tracking light, sound, motion, cut rate, cue timing, auditory envelope, or visual energy. The branch should not be called meaningless. A phase-scrambled branch can still produce visual entrainment, confusion, boredom, startle, sensory burden, artifact, or low-level affect. Its intended role is narrower: preserve sensory drive while damaging recognizable narrative meaning.

Control branch doctrine

Keep phase-scrambled Control for TSC-facing, publication-facing, OSF exemplar, and A-MRED confirmation-grade data. Use two-arm Target-vs-Override only for engineering tests, PRAYCG-lite demos, or private pilots with no meaning-vs-sensory-entrainment claim.

$\text{Control}(t) = \text{SensoryDrive}(t) + \text{CueEnergy}(t) + \text{ArtifactRisk}(t) - \text{RecognizableNarrative}(t)$

TargetSpecificMeaning requires: Target > Control after artifact, stimulus-fingerprint, and CET-R gates

Operationally, the Control branch is a meaning-damaged sensory entrainment control. It asks whether the response is just the physics of the stimulus, not whether the control is subjectively zero.

3. Decoder Gate Availability (DGA)

Eternal Sunshine forced a correction to the simple stimulus-transfer model. During that run, Target had the strongest absorption and afterglow but severe rain noise; Override had clearer audio but an analytic running-

sum burden. This means semantic access and emotional integration can come apart. DGA formalizes this dissociation.

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SA_b = 1 - mean(AudioComprehensionDifficulty, SpeakerVolumeDifficulty,
ExternalNoiseIntrusion, AudioVideoSyncProblem)/9
DA_b = mean(Meaning, Absorption, EmotionalAfterglow, StoryActiveWashout)/9
EI_b = mean(Absorption, EmotionalAfterglow, StoryActiveWashout)/9
X_b = max(TaskExtractionLoad_core, OverrideTaskBurden)/9
C_b = max(ConfoundBurden_core, DetailedConfoundMean)/9
D_gate,b = sigmoid(2.0*SA_b + 1.4*DA_b + 1.1*EI_b - 1.2*X_b - 1.8*C_b - 0.75)
GDI = mean(SA_Override-SA_Target, EI_Target-EI_Override, X_Override-X_Target, C_Target-
C_Override)
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DGA variable	Plain meaning	Why it matters
SA	Semantic access	A meaningful scene cannot be decoded if the audio/visual channel is unavailable.
DA	Decoder availability	The organism must be available to let a cue matter.
EI	Emotional/integrative impact	Target may affect the body even when semantic access is imperfect.
X	Extraction load	Override can preserve semantic recognition while taxing integration.
C	Confound burden	Noise, rain, sync problems, blur, and fatigue gate the endpoint.
GDI	Gate dissociation index	Captures split cases where semantic clarity and emotional integration separate.

4. MRED-Peak and MRED-Resolution

MRED has been compressed into two endpoint shapes. This does not remove the larger suite; it clarifies the interpretive hierarchy.

Endpoint	Question	Best archetype so far	Claim boundary
MRED-Peak	Did a locked anchor produce acute Target-specific meaning recognition plus delayed integration?	Contact father/avatar region	Requires locked anchors, QC, artifact pass, Control and Override contrasts.
MRED-Resolution	Did a meaningful event produce delayed reflective/regulatory recovery rather than a sharp peak?	Field of Dreams closure profile	Does not prove healing; Baseline2 is cumulative.
Recognition-dominant	Was meaning recognized without clean new integration?	Arrival	Not a failure; may represent familiarity or schema reactivation.
Decoder-gate dissociation	Did access, stance, and impact split across branches?	Eternal Sunshine	Confounded but theoretically informative.

$A-MRED_j = 1[MR_T > \theta_{MR}] * 1[ENC_T > \theta_{ENC}] * 1[T > C] * 1[T > O] * 1[QC = PASS]$

$MREDResolution_j = f(MR_T, RDI_T, Baseline2Echo, EET, Target > Control, Target > Override, QC)$

5. Narrative Update Polarity Index (NUPI)

NUPI asks not only whether meaning happened, but what kind of update the meaning demanded.

$ALI = \text{mean}(\text{semantic_intensity}, \text{Target_specificity}, \text{complexity_perturbation}, \text{TTI}, \text{primary_endpoint_pass})$

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RDI = mean(Baseline2_regulation, Baseline2_semantic_echo, EET_afterstate_echo,
self_report_echo)
NUPI = RDI - ALI

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Run	Current archetype	Interpretive status
Contact	High-load-with-recovery / MRED-Peak candidate	Strongest acute meaning-impact case, not pure crater.
Field of Dreams	Resolutive-recovery / MRED-Resolution candidate	Lower immediate intensity, stronger recovery tilt.
Arrival	Recognition-dominant / polarity unresolved	Useful for MR vs ENC theory; lacks PRAYCG2.0 Baseline2.
Eternal Sunshine	Decoder-gate dissociation	Target emotional impact and Override semantic clarity split by rain/audio confound.

6. Current evidence map: what may have been discovered

These are working hypotheses, not final discoveries in the confirmatory sense. They belong as a secondary part of the novelty section.

Working discovery	Evidence pattern	Boundary
Recognition and integration dissociate	Arrival and MRED tables motivate MR_HIGH/ENC_LOW or recognition-dominant cases.	Not proof of memory absence when theta is weak.
Peak and resolution meanings differ	Contact resembles acute peak; Field resembles delayed closure/recovery.	N=1 self-run archetypes; require replication.
Extraction reroutes meaning	Override can preserve semantic breakthrough while increasing task burden.	Not a moral score and not proof of total meaning erasure.
Sensory clarity and emotional impact are separable	Eternal Sunshine Target was more absorbing/afterglow; Override had clearer audio and was selected most meaningful.	Run is audio-confounded; useful as DGA prototype.
Meaning requires access conditions	DGA formalizes semantic access, decoder availability, integration, extraction load, and confounds.	Does not prove innate metaphysical state-space.
Phase-scrambled Control remains necessary	Control is often less narratively meaningful but physiologically nonzero.	Not proof that Control is meaningless.

7. Modern neuroscience interpretation: sensory entrainment versus meaning

The project should not be positioned against modern sensory-entrainment science. It depends on it. The phase-scrambled Control is the wall that lets PR-AYC-G ask whether anything remains after light, sound, pacing, cuts, cues, and rhythm have been modeled. The deeper hypothesis is that sensory entrainment is necessary but not sufficient for meaning integration. A brain can track a stimulus without receiving its meaning; a person can recognize meaning without integrating it; a person can be emotionally moved even when semantic access is degraded; and a person can understand more under Override while integrating less.

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SensoryTracking(t) = f(luminance, audio_envelope, optical_flow, cuts, cue_train)
MeaningIntegration(t+tau) = f(SensoryTracking_residual, D_gate, MR, ENC, Baseline2, EET)

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8. Current limitations and next confirmation path

- Most current runs are single-participant self-runs and therefore hypothesis-generating.
- Contact remains conceptually predeclared rather than runner-registered confirmation.

- Field of Dreams was runner-registered but estimated-anchor/timing-cautioned.
- Eternal Sunshine was operator-frame-verified but audio-confounded and ALS-cautioned.
- ALS timing must be improved with a fixed holder, longer or barcode pulse, and live pre-run pass/fail.
- Future confirmation-grade attempts should use frame-verified LOCKED anchors, clean ALS pulses across all branches, full StimulusFingerprint v1.8+, EOG/EMG when possible, and A-MRED / MRED-Peak-Resolution as the boxed primary endpoint.

Minimal confirmation-grade endpoint

A future result should be called confirmation-grade only when a frame-verified LOCKED anchor file is loaded before acquisition, ALS timing passes for Control/Target/Override, Target and Override hashes match, Control is generated from cue-embedded Target, StimulusFingerprint/CET-R files are available, artifact/confound gates pass, and A-MRED or MRED-Peak/Resolution passes under the predeclared criteria.

References to internal evidence used in this revision

Internal evidence note: *PRAYCG2.0 setup and run configuration; Field of Dreams and Eternal Sunshine self-reports/confounds; MediaPrep phase-scrambled control doctrine; Neurophenomenological self-report boundary; Structural-Epigenetic Memory Refresh theory; Opto-PING synthetic boundary.*